METRIC

MIL-DTL-38999/28G
30 May 2008
SUPERSEDING
MIL-DTL-38999/28F
11 June 2001

DETAIL SPECIFICATION SHEET

# CONNECTORS, ELECTRICAL, CIRCULAR, NUT, HEXAGON, CONNECTOR MOUNTING, SERIES III AND IV, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-38999.

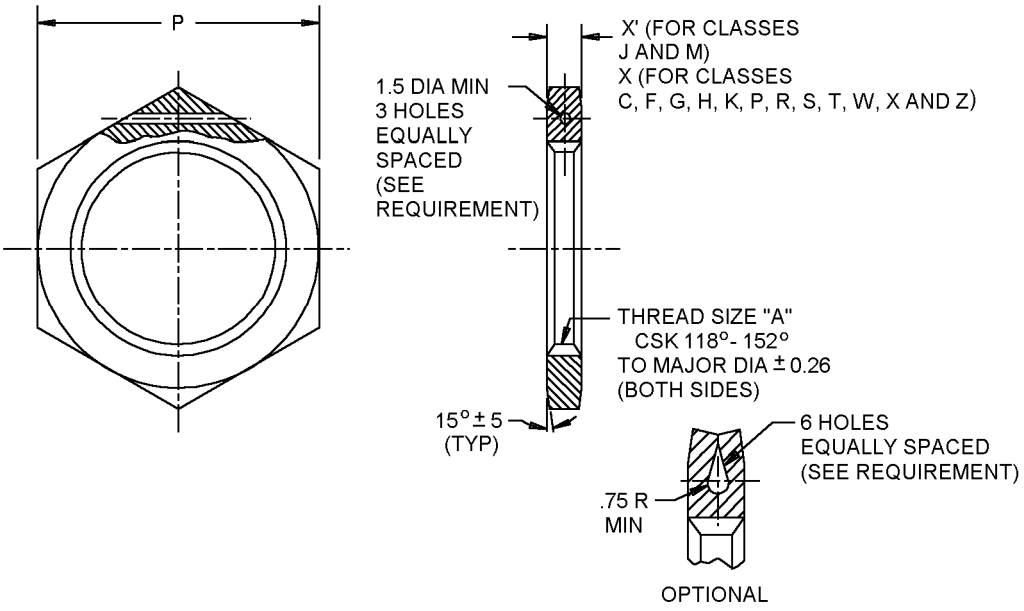



FIGURE 1. Nut, hexagon.



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| Dash number | Shell size | | A Thread size | P Hex | X' + .29 - .00 | X + .45 - .00 |
|---|---|---|---|---|---|---|
| | Series III and MIL-DTL-38999/43 of series IV | Series IV MIL-DTL-38999/44 only | | | | |
| 1 | 9 | -------- | M17X1 – 6H (see note 3) | 24.00 21.82 | 4.54 | 3.17 |
| 2 | 11 | -------- | M20X1– 6H (see note 4) | 27.00 24.99 | 4.54 | 3.17 |
| 3 | 13 | 11 | M25X1 – 6H | 32.00 29.77 | 4.54 | 3.17 |
| 4 | 15 | 13 | M28X1– 6H | 36.00 32.91 | 4.54 | 3.17 |
| 5 | -------- | 15 | M31X1 – 6H | 41.00 39.26 | 4.54 | 3.17 |
| 6 | 17 | -------- | M32X1 – 6H | 37.00 36.12 | 4.54 | 3.17 |
| 7 | ------- | 17 | M34X1 – 6H | 41.00 39.25 | 4.54 | 3.17 |
| 8 | 19 | ------- | M35X1 – 6H | 41.00 39.25 | 4.54 | 3.17 |
| 9 | 21 | 19 | M38X1 – 6H | 46.00 42.47 | 4.54 | 3.17 |
| 10 | 23 | 21 | M41X1 – 6H | 50.00 45.61 | 4.54 | 3.17 |
| 11 | 25 | 23 | M44X1 – 6H | 51.23 49.25 | 4.54 | 3.17 |
| 12 | ------- | 25 | M47X1 – 6H | 55.00 53.54 | 4.54 | 3.17 |

NOTES:
1. Dimensions are in millimeters.
2. Remove all burrs and sharp edges.
3. Modified minor diameter: 16.04 – 16.15.
4. Modified minor diameter: 19.07 – 19.15.

FIGURE 1. Nut, hexagon – Continued.



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REQUIREMENTS:

Dimensions and configurations: See figure 1.
Material: See table I.
Finish: See table I and MIL-DTL-38999.

TABLE I. Material and finish. 1/ 2/

| PIN code finish number | Material | Finish | Applicable class of connectors |
|---|---|---|---|
| C | Aluminum alloy | Anodic (nonconductive) | C |
| F | Aluminum alloy | Electroless nickel coating (conductive) | F, G, R |
| W | Aluminum alloy | Olive-drab cadmium plate over a suitable underplate (conductive) | W |
| X | Aluminum alloy | Same as W, but higher corrosion requirement | X |
| P | Aluminum alloy | Pure electrodeposited aluminum, non-reflective (conductive) | P |
| T | Aluminum alloy | Nickel fluorocarbon polymer, non-reflective (conductive) | T |
| Z | Aluminum alloy | Zinc nickel, non-reflective (conductive) | Z |
| K | Corrosion resistant steel | Passivated (conductive) | K, H, Y |
| N | Corrosion resistant steel | Electrodeposited nickel plating (conductive) | N, S, L |
| J | Resin with or without filler | Olive-drab cadmium plate over a suitable underplate (conductive) | J |
| M | Resin with or without filler | Electroless nickel coating, (conductive) | M |
| V | Resin with or without filler | Unplated (no finish) | J, M |

1/ See MIL-DTL-38999.
2/ Shall be cleaned, descaled and passivated in accordance with SAE-AMS2700.

Mating connectors: MIL-DTL-38999, series III and IV.

Safety wire hole pull out: Safety wire holes shall not pull out when safety wire NI-CU alloy (MONEL) (UNS NO4400) or NI-CR alloy (INCONEL) (UNS NO6600) 0.51 mm diameter in accordance with NASM20995 is threaded through a hole and a pull of 133.5 Newton minimum is applied. Three pulls shall be made, one parallel with axis of nut in each direction and one perpendicular to axis of nut.

Qualifications. Qualifications shall be in accordance with MIL-DTL-38999 except only the following tests shall be required:

Examination of product.
Safety wire hole pull out (see requirements).
Salt spray (corrosion).



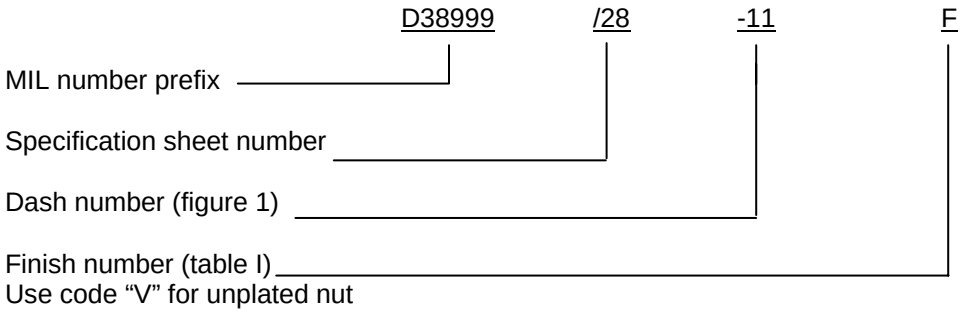
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Group A sampling inspection. Sampling for group A inspection shall be specified in table II.

TABLE II. Group A sampling plan.

| Lot size | Sampling size |
|---|---|
| 1 to 13 | 100 percent |
| 14 to 150 | 13 units |
| 151 to 280 | 20 units |
| 281 to 500 | 29 units |
| 501 to 1,200 | 34 units |
| 1,201 to 3,200 | 42 units |
| 3,201 and up | 54 units |

Part or Identifying Number (PIN) example:




NOTE: The term PIN is equivalent to the term (part number, identification number, and type designator) which was previously referenced in this specification.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-38999, this document references the following:

MIL-DTL-38999/43
MIL-DTL-38999/44
NASM20995
SAE-AMS2700



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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy - AS
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2007-137)

Review activities:
Army - AR, MI
Navy - EC, MC, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change. You should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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